

Tanvir Hossain Ovi

✉ tanvirhossainovi.eee@std.cu.ac.bd ☎ +880 1863 587030 📍 Chattogram, Bangladesh
🌐 tanvir-hossain-ovi.me 🔗 LinkedIn 🐙 GitHub 🎓 Google Scholar

EDUCATION

University of Chittagong, Bangladesh

Jan 2020 – Nov 2025

BSc in Electrical and Electronic Engineering | **Engineering Degree with Honours**

CGPA: 3.78 / 4.00 | **Merit Position: 2nd out of 63 students**

Thesis: *EEG-based cognitive load analysis in 360° virtual reality and 2D laptop video modes*

RESEARCH INTERESTS

Brain-computer interfaces • Neural decoding • EEG/iEEG signal analysis • Cross-subject generalization

PUBLICATIONS

Nine papers: three published journal articles, one accepted IEEE conference paper, and five under review or submitted. First author on five.

Journal Articles

- [P1] **T. H. Ovi**, M. M. Islam, M. A. Uddin, K. Ghazinour, A. M. M. Chowdhury, and N. Siddique, “Frequency Band Dissociation Between Cognitive State Recognition and Person Identification in EEG,” *Machine Learning with Applications*, p. 100901, 2026.
- [P2] M. A. Z. Akas, **T. H. Ovi**, and M. F. Kader, “Impact of Mobile Phone Use on the Brain Activity: Audio Call vs Video Call,” *Acta Psychologica*, vol. 262, p. 106160, 2026.
- [P3] A. Hossain, **T. H. Ovi**, M. A. I. Mahmood, and M. F. Kader, “Classification of Envisioned English Speech from EEG using Deep Learning Approaches,” *Machine Learning with Applications*, vol. 22, p. 100752, 2025.

Conference Paper

- [P4] **T. H. Ovi**, A. M. M. Chowdhury, K. Ghazinour, and M. Simsek, “Design of a Clustering-Enabled Multi-Stage Intrusion Detection Framework for APT Mitigation in IoT Networks,” in *Proceedings of the IEEE Conference on Dependable and Secure Computing (DSC)*, 2026. [Accepted]

Under Review and Submitted

- [P5] **T. H. Ovi**, T. Forsythe, A. M. M. Chowdhury, N. Siddique, and Y. Li, “Phenotype-Aware Personalization for Cross-Session EEG-Based Emotion Recognition,” *Expert Systems With Applications*. [Under Review]
- [P6] A. Hossain, M. A. Z. Akas, **T. H. Ovi**, and M. F. Kader, “Heterogeneous Deep Learning Approaches for EEG-Based Imagined Bangla Word-Level Speech Recognition,” *Computer Speech & Language*. [Under Review]
- [P7] **T. H. Ovi**, A. M. M. Chowdhury, K. Watanabe, and N. Siddique, “Complementary Raw and Engineered Representations for Focal and Non-Focal Intracranial EEG Classification under a Leakage-Controlled Recording-Level Protocol,” *Neurocomputing*. [Submitted]
- [P8] **T. H. Ovi**, T. F. Arin, M. A. I. Mahmood, and M. F. Kader, “EEG-Based Cognitive Load Analysis in Immersive Virtual Reality and Laptop-Based Video Learning Environments,” *Applied Cognitive Psychology*. [Submitted]
- [P9] T. F. Arin, **T. H. Ovi**, and M. F. Kader, “Bangla Imagined Speech Using EEG to Understand Emotional Words,” *Acta Psychologica*. [Submitted]

RESEARCH EXPERIENCE

Research Assistant, Biometric Intelligence and Vision Laboratory Jul 2025 – Present
Department of Cybersecurity, State University of New York, Canton, USA | Remote
Supervised by Dr. A M Mahmud Chowdhury and co-supervised by Dr. Nazmul Siddique (Ulster University, UK). First author on all four studies from the appointment [P1, P4, P5, P7].

- **Intracranial EEG classification** [P7]. Traced inflated Bern-Barcelona accuracies to window-level splits, rebuilt the evaluation at recording level, and reached 96.27% with CSRA-EF Net.
- **Cross-session emotion recognition** [P5]. Placed 80.1% of explained variance in the participant-by-emotion interaction. Phenotype calibration raised accuracy from 39.96% to 88.71%.
- **Frequency band dissociation** [P1]. Delta EEG reached 86.9% task accuracy but only 15.6 to 23.6% authentication, while broadband reached 69.9%.

Undergraduate Researcher, University of Chittagong 2023 – 2025
Department of Electrical and Electronic Engineering, Chattogram, Bangladesh

- Recorded EEG from participants for four studies: the VR thesis, imagined Bangla words, emotionally valenced Bangla words, and audio vs. video calls [P2, P6, P8, P9].
- **BSc thesis** [P8]. Designed a VR vs. laptop video learning protocol and recorded 30 participants on a 14-channel Emotiv EPOC X. VR lowered cognitive load by 21% and raised engagement by 59%.

HONORS AND AWARDS

- First Runner-up, Student Merit Award, EEE Fest, University of Chittagong Feb 2025
- Champion, Robo Soccer Competition, EEE Fest, University of Chittagong Jan 2023
- Second Runner-up, Robo Soccer Competition, Engineering Day, University of Chittagong Jun 2022

TEST SCORES

IELTS Academic 14 April 2026
Overall Band Score: **7.5** | Listening: 8.0 Reading: 8.0 Writing: 6.0 Speaking: 7.0

TECHNICAL SKILLS

Programming Python, PyTorch, TensorFlow, Keras, scikit-learn, Matplotlib, Git, Linux, LaTeX
EEG Emotiv EPOC X recording, MNE-Python, EEGLAB, ICA artifact removal
Statistics t-test, Wilcoxon, ANOVA, MANOVA, FDR correction, hierarchical clustering

REFERENCES

Dr. Nazmul Siddique Senior Lecturer School of Computing, Engineering and Intelligent Systems Ulster University, UK <i>Research Supervisor</i> nh.siddique@ulster.ac.uk	Dr. A M Mahmud Chowdhury Assistant Professor Department of Cybersecurity SUNY Canton, USA <i>Research Supervisor</i> Chowdhuryam@canton.edu	Dr. Kazi Tanvir Ahmmed Professor Department of Electrical and Electronic Engineering University of Chittagong <i>Head of Department</i> tanvir@cu.ac.bd
--	---	---